Employee Management System – Functional Test Cases

Programming Task A – Update Employee Data

Description: Implement functionality to update any editable employee information (name, email, phone, salary, division, job title, etc.) using the Edit Employee / Employee Form screens. Changes must be persisted in the employees table and related mapping tables (employee_division, employee_job_title).

Test Case ID: 1.1
Description: Update an existing employee with valid data (happy path).
Test Inputs:
- Employee ID: 105 (existing)
- New Name: "Alex Johnson"
- New Email: "alex.johnson@example.com"
- New Division: "Engineering"
- New Job Title: "Senior Developer"
- New Salary: 82000

Expected Results (Oracle):
- Employee 105 record in the employees table is updated with the new name and email.
- The employee_division mapping points to the Engineering division.
- The employee_job_title mapping points to "Senior Developer".
- The latest payroll record reflects salary 82000 (or a new payroll row is created according to business
rules).

Dependencies:
- Employee with ID 105 exists in the database with valid division and job title.
- Divisions and job titles used in the test are already configured in lookup tables.

Initialization:
- Start the application and navigate to the main menu.
- Verify database is in a known state and employee 105 exists with original data recorded for comparison.

Test Steps:
1. From the main menu, open the Edit Employee screen.
2. Search for employee 105 by Employee ID and load their current details.
3. Change the fields to the values listed in Test Inputs.
4. Click the Save or Update button.
5. Re-open employee 105 (or query the DB) and verify all updated fields match the Test Inputs.

Tear Down:
- Optionally revert employee 105 to original data if the environment must be reset.

Test Case ID: 1.2
Description: Attempt to update a non-existent employee.
Test Inputs:

- Employee ID: 9999 (does not exist)
- Any updated field values (name, email, etc.)

Expected Results (Oracle):
- System displays an error message such as "Employee not found".
- No rows are inserted or updated in employees or related tables.

Dependencies:- There is no employee with ID 9999 in the database.

Initialization:
- Application running; main menu accessible.

Test Steps:
1. Open the Edit Employee screen.
2. Enter Employee ID 9999 in the search field and attempt to load details.
3. Verify that the system shows an appropriate error message and does not allow editing.

Tear Down:
- None (database should remain unchanged).

Test Case ID: 1.3
Description: Attempt to update an employee with missing required fields (validation failure).
Test Inputs:
- Employee ID: 105 (existing)
- Name: "" (left blank)
- All other fields valid

Expected Results (Oracle):
- Validation error is displayed indicating that Name is required.
- Save/Update is blocked.
- No changes are persisted to the database.

Dependencies:
- Employee 105 exists.

Initialization:
- Application running; database in known state.

Test Steps:
1. Load employee 105 in Edit Employee screen.
2. Clear the Name field and leave other fields unchanged or valid.
3. Click Save/Update.
4. Verify that a validation message appears and that employee data in the DB remains unchanged.

Tear Down:
- None (no committed updates expected).

Programming Task B – Search for Employee

Description: Implement search functionality that allows users to find employees by Employee ID, SSN, or Name. Results should be shown in the Search and Edit screens and must handle cases with zero, one, or many matches.

Test Case ID: 2.1
Description: Search for an existing employee by Employee ID (happy path).

Test Inputs:
- Search Mode: Employee ID
- Query: 102 (existing employee)

Expected Results (Oracle):
- Employee with ID 102 is displayed with correct name, SSN, division, and job title.

Dependencies:
- Employee 102 exists with known details.

Initialization:
- Application running; database seeded.

Test Steps:
1. Navigate to the Employee Search screen.
2. Select "Employee ID" in the search mode dropdown.
3. Enter 102 in the search field and click Search.
4. Verify that a single row is displayed with the correct data for employee 102.

Tear Down:
- None.

Test Case ID: 2.2

Description: Search for an existing employee by SSN (happy path).

Test Inputs:
- Search Mode: SSN
- Query: 123456789 (SSN for an existing employee)

Expected Results (Oracle):
- The employee with SSN 123456789 is displayed.

Dependencies:
- An employee exists with SSN 123456789 in the employees table.

Initialization:
- Application running; test SSN known ahead of time.

Test Steps:
1. Open Employee Search screen.
2. Select "SSN" as search mode.
3. Enter 123456789 and click Search.
4. Verify that exactly one matching employee is shown with correct data.

Tear Down:
- None.

Test Case ID: 2.3
Description: Search by partial name returning multiple employees (pass scenario).

Test Inputs:
- Search Mode: Name
- Query: "Ann" (substring shared by multiple employees)

Expected Results (Oracle):
- A list of all employees whose names contain "Ann" (case-insensitive) is returned.
- Each row displays the correct ID, name, and other basic info.
Dependencies:- At least two employees with "Ann" in their name exist in test data.
Initialization:
- Application running; database seeded appropriately.

Test Steps:
1. Open Employee Search screen.
2. Select "Name" mode.
3. Enter "Ann" and click Search.
4. Verify that multiple rows are displayed and each matches an employee with "Ann" in the name.

Tear Down:
- None.

Test Case ID: 2.4
Description: Search for a non-existing employee (fail path – no results).
Test Inputs:
- Search Mode: Name
- Query: "qwertyzzz" (name not present in DB)
Expected Results (Oracle):
- No employees are returned.
- The user sees a "No results found" message or an empty table with a suitable status message.
Dependencies:
- No employee in the database contains "qwertyzzz" in their name.
Initialization:
- Application running.

Test Steps:
1. Open the Employee Search screen and select "Name" mode.
2. Enter "qwertyzzz" and click Search.
3. Verify that zero results are displayed and a friendly message is shown.

Tear Down:

- None.

Test Case ID: 2.5
Description: Search by Employee ID with invalid (non-numeric) input (validation failure).
Test Inputs:
- Search Mode: Employee ID
- Query: "ABC"
Expected Results (Oracle):
- System displays a validation error such as "Employee ID must be a number".
- No search is executed against the database.
Dependencies:
- None (pure input validation).
Initialization:
- Application running.
Test Steps:
1. Open Employee Search screen and select "Employee ID" mode.
2. Enter "ABC" and click Search.
3. Verify that a validation message is shown and that no results table is refreshed from the DB.
Tear Down:
- None.

Programming Task C – Update Salary for Employees Below a Particular Amount

Description: Implement a salary adjustment feature that increases the salary (payroll amount) of all employees whose current salary is below or within a specified threshold/range by a given percentage. The system must show how many employees were affected and maintain pay history.

Test Case ID: 3.1
Description: Apply a valid percentage increase to all employees with salaries between 5,000 and 10,000 (happy path).
Test Inputs:
- Min Salary: 5000
- Max Salary: 10000
- Increase Percentage: 5%
Expected Results (Oracle):
- All employees whose latest payroll amount is between 5,000 and 10,000 receive a 5% increase.
- For each affected employee, a new payroll record is created with the increased amount and correct pay period dates.
- A summary displays the count of employees updated and possibly the total additional monthly cost.
Dependencies:
- Database seeded with employees having salaries in and out of the range.
Initialization:
- Application running; original payroll data recorded for comparison.
Test Steps:
1. Open the Salary Adjustment screen.

2. Enter Min Salary = 5000, Max Salary = 10000, and Percentage = 5.
3. Click Apply / Update.
4. Verify that the confirmation shows at least one employee updated.
5. Query payroll for affected employees and ensure new records have amount = old_amount * 1.05.

Tear Down:
- Optionally roll back payroll data to original values.

Test Case ID: 3.2
Description: Apply a percentage increase when no employees fall under the specified threshold (logical fail case).
Test Inputs:
- Min Salary: 10000
- Max Salary: 15000
- Increase Percentage: 10%

Expected Results (Oracle):
- System completes the operation but reports "0 employees affected".
- No new payroll records are inserted; existing payroll data remains unchanged.

Dependencies:
- No employees currently have salaries between 10,000 and 15,000 in test dataset.

Initialization:
- Application running; DB verified to have no salaries in this range.

Test Steps:
1. Open Salary Adjustment screen.
2. Enter Min Salary = 10000, Max Salary = 15000, Percentage = 10.
3. Click Apply / Update.
4. Verify that the UI reports 0 employees updated and DB data is unchanged.

Tear Down:- None.

Test Case ID: 3.3
Description: Attempt salary update with invalid range (Min greater than Max).
Test Inputs:
- Min Salary: 90000
- Max Salary: 30000
- Increase Percentage: 5%

Expected Results (Oracle):
- System detects invalid range and shows validation message such as "Minimum salary must be less than maximum salary".
- No salary update is processed and no payroll data is changed.

Dependencies:
- None (validation only).

Initialization:
- Application running.

Test Steps:
1. Open Salary Adjustment screen.
2. Enter Min Salary = 90000 and Max Salary = 30000 with Percentage = 5.
3. Click Apply / Update.
4. Verify that the user sees range validation error and no DB changes are made.

Tear Down:
- None.

Test Case ID: 3.4
Description: Attempt salary update with invalid percentage (negative or zero value).

Test Inputs:
- Min Salary: 0
- Max Salary: 60000
- Increase Percentage: -3% (or 0%)

Expected Results (Oracle):
- The system displays validation error such as "Increase percentage must be greater than 0".
- No payroll data is modified.

Dependencies:
- None.

Initialization:
- Application running.

Test Steps:
1. Open Salary Adjustment screen.
2. Enter Min Salary = 0, Max Salary = 60000, Percentage = -3 (or 0).
3. Click Apply / Update.
4. Verify that a validation message is shown and that no payroll rows are changed.

Tear Down:
- None.